

Estimating measures to reduce the transmission of SARS-CoV-2 in Australia to guide a ‘National Plan’ to reopening

Gerard E. Ryan^{1,2}, Freya M. Shearer², James M. McCaw^{2,3}, Jodie McVernon^{4,5}, and Nick Golding^{1,6}

¹Telethon Kids Institute, Nedlands, WA, Australia

²Centre for Epidemiology and Biostatistics, Melbourne School of Population and Global Health, The University of Melbourne, VIC, Australia

³School of Mathematics and Statistics, The University of Melbourne, Melbourne, Australia

⁴Department of Infectious Diseases, Melbourne Medical School, University of Melbourne at the Peter Doherty Institute for Infection and Immunity, Melbourne, Australia

⁵Victorian Infectious Diseases Reference Laboratory, The Royal Melbourne Hospital at the Peter Doherty Institute for Infection and Immunity, Melbourne, Australia

⁶Curtin University, Perth, Australia

ABSTRACT

Background

In mid-2021, widespread availability of COVID-19 vaccines with demonstrated impacts on transmission promised relief from the strict public health and social measures (PHSMs) imposed in many countries to limit spread and burden. We were asked to define vaccine coverage thresholds for transition through the stages of Australia’s ‘National Plan’ to easing restrictions and reopening international borders.

Methods

Using available evidence of vaccine effectiveness against the then-circulating Delta variant, we used a mathematical model to determine vaccine coverage targets. The absence of any COVID-19 infections in many sub-national jurisdictions in Australia posed particular methodological challenges for modelling in this setting. We used a novel metric called Transmission Potential (TP) as a proxy measure of the population-level effective reproduction number. We estimated TP of the Delta variant under a range of PHSMs, test-trace-isolate-quarantine (TTIQ) efficiencies, vaccination coverage thresholds, and age-based vaccine allocation strategies.

Findings

We found that high coverage of vaccination across all age groups ($\geq 70\%$) combined with ongoing TTIQ and minimal PHSMs was sufficient to avoid strict lockdowns. At lesser coverage ($\leq 60\%$) rapid case escalation risked overwhelming of the health sector and would prompt a need to reimpose strict restrictions, with substantive economic impacts in order to achieve the goals of the National Plan. Maintaining low case numbers was the most beneficial strategy for health and the economy, and at higher coverage levels ($\geq 80\%$) further easing of restrictions was deemed possible.

Interpretation

These results reinforced recommendations from other modelling groups that some level of PHSMs should be continued to minimise the burden of the Delta variant following achievement of high population vaccine coverage. They directly informed easing of COVID-19 restrictions in Australia.

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Keywords: COVID-19, Australia, transmission potential, vaccination, public health, intervention, economic modelling

INTRODUCTION

Since early 2020, rapid dissemination of SARS-CoV-2 and subsequent emergence of variants have resulted in multiple rapidly escalating waves of infection with devastating impacts on health, society and the economy [1]. In contrast with the rest of the world, many island nations in the Western Pacific Region remained relatively COVID free through the first two years of the pandemic as a result of strong border controls to prevent importation, and reactive imposition of social restrictions to constrain community transmission (e.g. [2, 3]). These measures enabled enviable social freedoms and ongoing economic activity, but such disconnection from the international community was not a sustainable strategy over the longer term [4].

Global concerted efforts to accelerate development and licensure of safe and effective vaccines raised hopes that wide scale population immunisation would enable a return to ‘life as normal’ in high burden settings, given demonstrated impacts on infection acquisition and onward spread [5, 6, 7]. For countries that had pursued a low or zero-COVID strategy, ‘living with COVID’ seemed an achievable goal if vaccines could constrain transmission and mitigate disease outcomes sufficiently to avoid overwhelming the health system following SARS-CoV-2 importation, while maintaining near normal societal functioning [8, 9]. However, anticipating the likely impacts of introducing COVID-19 into an environment without established SARS-CoV-2 transmission is difficult [10].

To address this challenge, we developed a Transmission Potential (TP) [10] metric to quantify the likely spread of COVID in populations based on behavioural, social mixing data and available case data. TP is an estimate of the average population-level trend in the effective reproduction number (R_{eff}) and is driven by factors that influence transmission from local cases. Relevant considerations include the effectiveness of test-trace-isolate-quarantine (TTIQ) case and contact management [11], as well as changes in personal distancing behaviour and mandated constraints on mixing group sizes, collectively termed public health and social measures (PHSMs) [10]. Demonstrated reductions in TP achieved by these interventions can be incorporated into future scenario projections, along with anticipated impacts of vaccination on transmission [REF vaccination efficacy/TP paper]. International evidence of vaccine effectiveness against acquisition and onward spread of the Delta variant was used to estimate the likely overlaid impact of differing levels of vaccine coverage (by age cohort) on population wide TP.

This approach was used in mid-2021 to help inform Australia’s National Plan to transition Australia’s National COVID-19 response[12], by determining target vaccination thresholds for moving between its four phases (table 1), including any ongoing requirement for public health responses and social measures. These methods formed the basis for further collaborative work with the Australian Government Treasury to determine likely economic implications of reopening at alternative thresholds based on the level of adjunct PHSMs needed [13], and constituted part of a broader suite of modelling advice to the Australian Government to inform management of the pandemic and a shift away from strict lockdown measures [14, 15, 16]. From the outset, it was anticipated that reopening targets in populations naïve to COVID-19 would be higher than for those with a history of prior circulation resulting in some degree of established immunity to the virus.

METHODS

Transmission Potential

Transmission potential (TP) is a measure of the average potential for a virus to spread at the population level [10]. It can be considered as an estimate of the reproduction number, if transmission were widespread (and therefore not concentrated in one demographic group with non-representative transmission rates). The base model consists of three sub-models involving time to detection and isolation of cases, and two types of physical distancing behaviour: we differentiate “macro-distancing” as the reduction in the average rate of non-household contacts (e.g. in response to lockdown-type restrictions), while “micro-distancing” is the reduction in transmission probability per non-household contact (e.g. adherence to social-distancing and hygiene advice or legislation; [10]). Transmission potential will vary among communities by differential average household size and age structures, and can be modulated by changes to behaviour through public health and social measures, time to detection and isolation of cases, and immunity through infection and vaccination. A full description of this model can be found in Golding and colleagues [10]. Whilst an

Table 1. Phases of the “National Plan to transition Australia’s National COVID-19 Response”[12]. Our modelling analysis focuses on the transition from ‘phase A’ in which strong suppression and no community transmission is the goal, to ‘phase B’ where vaccine coverage is high and SARS-CoV-2 infection is allowed to establish in the population. Scenarios therefore examine the epidemic dynamics and clinical consequences of infections following seeding of an epidemic at different vaccination coverage thresholds achieved through alternative age prioritisation strategies.

Phase	Description	Activities
A	Vaccinate, Prepare and Pilot	Continue to strongly suppress the virus for the purpose of minimising community transmission
B	Vaccination Transition	Seek to minimise serious illness, hospitalisation and fatality as a result of COVID-19
C	Vaccination Consolidation	As above
D	Post-Vaccination	Manage COVID-19 consistent with public health management of other infectious diseases

estimate of transmission potential under a given scenario of R_0 , behaviour, and health system performance can never be a perfect predictor of a reproduction number due to epidemic stochasticity and unaccounted sources of variation, it is a measure that enables prediction of the likely ability to control spread in the event of widespread transmission, and has been used as a systematically reported situational assessment metric in Australia since 2020 [17, 18, 19].

Test, Trace, Isolate, and Quarantine

Test-trace-isolate-quarantine (TTIQ) strategies are key non-pharmaceutical interventions used globally to manage infectious disease outbreaks [20, 21] including frequently during the COVID-19 pandemic [22, 23]. TTIQ operates through isolating cases identified by testing, and tracing and pre-emptive quarantining of their close contacts to prevent further onward transmission. We adapted our TP model to include an explicit effect of reducing the time to case isolation achievable through intensive contact tracing, in addition to the time to case detection effect already included [11]. The empirical distribution of times from symptom onset to case isolation under an ‘optimal’ TTIQ capacity (i.e. with a health system that had enough capacity to rapidly contact trace all cases) was estimated using a limited time series of case data from the state of New South Wales between July 2020 and January 2021, for which dates of isolation were known with a high level of data completeness. This distribution was then calibrated to estimate the distribution of times to isolation in other times and states by assuming improvements in TTIQ are proportional to improvements in times to detection. We characterised a second level of ‘partially’ efficacious TTIQ based on observations from the state of Victoria on 4 August 2020. This date was the then-peak of daily locally-acquired COVID-19 cases in Australia representing a contact-tracing system under resource constraints. These data were used to estimate the effect of ‘partial’ TTIQ on transmission potential to estimate a baseline TP under community transmission. These estimates of ‘optimal’ and ‘partial’ TTIQ correspond to a 54% and 42% reduction in transmission respectively; the full details of this estimation are found in [11].

Public Health and Social Measures

In the presence of ongoing viral transmission, it is necessary to keep the rate of virus reproduction, R_{eff} , below or close to 1, as any extended periods where $R_{eff} > 1$ can quickly lead to significant numbers of cases, causing stressors on health systems. So maintenance of R_{eff} near or below 1 is needed both to contain community transmission in the suppression phase, and to prevent cases from exceeding health sector capacity after re-opening. Regulated or advised risk reduction behaviours and constraints on social mixing described collectively as public health and social measures (PHSMs) are the levers that may be employed to manage R_{eff} in response to incursions and outbreaks. Behaviours change over time either spontaneously because of heightened concern or complacency, or in response to mandated public health orders invoking various elements of PHSMs.

Here we investigated what level of PHSMs would be required to bring TP to near or below 1 under different scenarios of vaccination coverage. In collaboration with the Australian Government Treasury we defined four ‘bundles’ of PHSM restrictions: baseline, low, medium and high. Each bundle is drawn

from a specific time and place in Australia's pandemic experience, thereby capturing both real-world behavioural responses and the proportional reduction in TP achievable by PHSMs in this context. They therefore represent differing degrees of social constraints for which the impact on TP and economic activity could be defined from historical observations.

Full descriptions of measures included in these bundles are in table S1. These bundles are intended to reflect the plausible behavioural consequences of different levels of stringency of PHSM. As such, the 'Baseline PHSM' bundle reflects behaviours during a period with minimal restrictions, and a population aware of the risk of COVID-19 (as opposed to behaviour prior to the pandemic). The other bundles reflect the behavioural effects of increasing levels of stringency. We emphasise that the TPs associated with these PHSM bundles reflect state wide population behaviours (numbers of household contacts and adherence to hygiene advice) estimated at these times. TPs are observed to differ substantially over time and between states, even within similar restrictions [10, 18]. These periods are therefore intended to reflect achievable levels of reduction in TP via PHSMs, rather than inference about the particular impacts of these sets of restrictions.

Table 2. Description of measures implemented under different 'bundles' of public health and social measures (PHSMs). Each bundle relates to a specific time and place in Australia's pandemic experience up to mid-2021 — thereby capturing behavioural responses and the proportional reduction in TP achievable by PHSMs in the Australian context. The proportional reductions in TP observed at each time and place can therefore be related to similar reductions achieved via other combinations of PHSMs (not limited to the "bundles" in place during the reference period). Similarly, the imposition of any given combination of PHSMs at different times and places may result in variable population responses and thus reductions in TP. More detailed descriptions of the bundles can be found in S1

PHSM bundle	Description
Baseline	Minimal density/capacity restrictions and no major outbreaks as in NSW March 2021
Low	More stringent capacity restrictions compared to baseline (<i>e.g.</i> , hospitality venues limited to 10 customers per booking), as in NSW 23 August 2020
Medium	Stringent capacity restrictions, group size limits (<i>e.g.</i> , fewer than five people outdoors), stay-at-home orders (except work, study, essential purposes), as in NSW 1 July 2021
High	No household visitors, curfew, stay-at-home orders (except essential purposes and permitted work), schools closed (remote learning only), as in VIC 23 August 2020

Age-structured Vaccination Impact on Transmission

In keeping with WHO guidance [24] and many other countries, Australia's immunisation program initially prioritised health- and aged-care workers, elderly populations and those at increased risk of transmission and/or severe outcomes [25]. When defining overall target coverage thresholds for the eligible population, an important goal was to consider the distribution of doses received across age categories. While older individuals are more likely to experience severe disease outcomes, young and working adults are expected to make a greater contribution to transmission at the population level. So to explore the effect of age-structured vaccination on transmission, we explored four allocation strategies: oldest first, 40+ years first, all adults, and transmission reducing, described in table 3 (the specific percentages of each age-group vaccinated in these strategies can be found in [16]). These strategies were motivated by being the strategy in place at the time (oldest first), and a series of what were considered the only other practical options. The strategies followed the Australian Therapeutic Goods Administration [26, 27] and Australian Technical Advisory Group on Immunisation [28] advice on which vaccines were administered to which age-groups and the dosing interval between first and second doses 4. We also assume a two week delay from administration until a vaccine takes full effect 4.

We estimated the percentage reduction in TP that could be expected under different vaccination coverages and distributions by age, vaccine type, and the number of doses received, via static analysis of an age-based transmission matrix, S2, [29]. Age-specific susceptibility and transmissibility estimates [30] are used and transmission rates have been calibrated to a baseline population-wide TP of 3.6. TP will be

influenced by spontaneous and imposed changes in physical distancing behaviours, the number of social contacts on average between individuals and the timeliness of TTIQ measures. We use a baseline TP of 3.6 for the Delta variant based on averaged observations from the state of New South Wales in March 2021, a period with minimal social restrictions and no major outbreaks.

For each vaccination scenario, the reduction in transmission by age-group was calculated from the average vaccination efficacy against transmission (accounting for the fractions of each vaccine type and number of doses in that age-group) and the age-group coverage. Proportional reductions in transmission rates for each age combination were then applied to the original ‘unvaccinated’ transmission matrix to construct a ‘vaccinated’ matrix. The dominant eigenvalue, representing the population-wide reproduction number was compared between these pairs of matrices to compute a percentage reduction in TP due to immunisation.

To explore and visualise the effect of different strategies on TP reduction among age categories, we also calculated an age-group specific TP prior to and after vaccination under a given scenario. These ‘by-age’ contributions are calculated for a given homogeneous age-group (whereas other TP calculations use all age groups concurrently S2). Because these age-specific TP calculations exclude interactions with other age-groups, they are not equivalent to the partial contribution of that age group to the overall TP.

Table 3. Vaccine allocation strategies and allocation sequence by age-group. The specific modelled percentages of each age-group vaccinated in these strategies can be found in [16].

Strategy	Allocation sequence
Oldest first	Vaccinations are prioritised from oldest to youngest. Specifically, prioritization occurs in the following order: 80+, 70-79, 60-69, 50-59, 40-49, 30-39, 20-29, 16-19
40+ years first	Vaccinations are prioritised from 40+ upwards, then 16+. Specifically, prioritization occurs in the following order: 40-49, 50-59, 60-69, 70-79, 80+, 16-19, 20-29, 30-39
All adults	Vaccinations are not prioritised in any particular order by age
Transmission reducing	As for national program, under which all individuals 40+ were eligible as at the 12 th of July 2021. Within the simulation time-frame, the 30-39 years cohort becomes eligible from the 30 th of August 2021, and 16-29 years on the 11 th of October.

Time in Lockdown

During outbreak suppression in Australia, early and stringent lockdowns were used to bring TP below 1 for the purposes of driving even a handful of local cases from an outbreak to zero, in the context of an optimal TTIQ response. The goal of transitioning to phase B 1 was to minimise the requirement for such stringent PHSMs, restricting their use to meet the explicit objective of prevention of overwhelming the health sector in the face of escalating case loads.

Ongoing application of some degree of social measures through this phase to support vaccine impacts reduces the likelihood for high restrictions and preserves TTIQ effectiveness by keeping case numbers low. TP estimates with and without stringent PHSM can be used to calculate the approximate proportion of time those stringent measures would need to be in place to prevent exceedance of health sector capacity over a defined time frame. This simple static analysis can indicate the plausible societal and economic impacts of the PHSM required to constrain transmission under each scenario and coverage.

Where a vaccination scenario leads to either a $TP_1 > 1$ with one PHSM bundle and $TP_2 < 1$ with a more stringent bundle, the long-term average TP can be maintained at 1, i.e. with daily case counts neither growing nor shrinking over the long term, by alternating between the two PHSM bundle states.

Table 4. Vaccine eligibility, dosing intervals, and assumed delay to efficacy.

Vaccine	Eligible population	Dosing interval	Delay to full efficacy
Vaxzevria (ChAdOx1-S) (AZ) [27]	60+ years [28]	12 weeks	2 weeks
Comirnaty (BNT162b2) (Pf) [26]	16+ years	3 weeks	2 weeks

Whilst the first PHSM bundle is in place cases will grow, and whilst the more stringent bundle is in place cases will shrink, leading to an oscillation of case counts around some average level S_1 . This fraction of time under more stringent PHSMs is independent of the sequence or duration of the periods under more stringent restrictions; a strategy of rapid switching on and off of restrictions, or one of alternating long periods with or without restrictions would both lead to long-term maintenance of $TP = 1$, provided the fraction of time in each condition is the same S_1 .

Switching between more and less stringent PHSMs reflects a strategy that might be used to keep cases below a health sector capacity limit in the event that there is long-term community transmission. With the necessary simplifying assumption that vaccination coverage is static, where one PHSM bundle leads to growing cases numbers ($TP_1 > 1$), and a second bundle leads to contracting case numbers ($TP_2 < 1$), we can calculate the fraction of time necessary under each bundle of PHSMs S_1 . However this strategy will not always be either necessary or possible. When cases under both bundles lead to declining case numbers (i.e. $TP_1 < 1$ and $TP_2 < 1$), the fraction is zero as the more stringent PHSM bundle is not needed. Alternatively, where even the more stringent PHSMs still lead to growing daily case numbers ($TP_2 > 1$) no fraction exists, because even the more stringent PHSM bundle could not control transmission.

Costs of PHSMs

The Australian Government Treasury estimated the direct economic costs of alternative COVID-19 management scenarios explored in this analysis. Estimates included the expected average weekly costs of activity restrictions and lockdowns for each of the bundled levels of restrictions, multiplied by their duration of application over the specified timeframe. These figures were derived by analysing the impact on hours worked across the economy during lockdown periods in 2020, compared with the pre-COVID baseline. They did not include indirect confidence effects, labour market impacts, social, fiscal or health economic costs. For all scenarios it was assumed that case numbers would be constrained by social measures to avoid overwhelming the health system. An objective of maintaining low case numbers in this way was to avoid the significant behavioural changes and related economic impacts that were observed in other country settings where severe and widespread outbreaks occurred in the absence of mixing restrictions. [13].

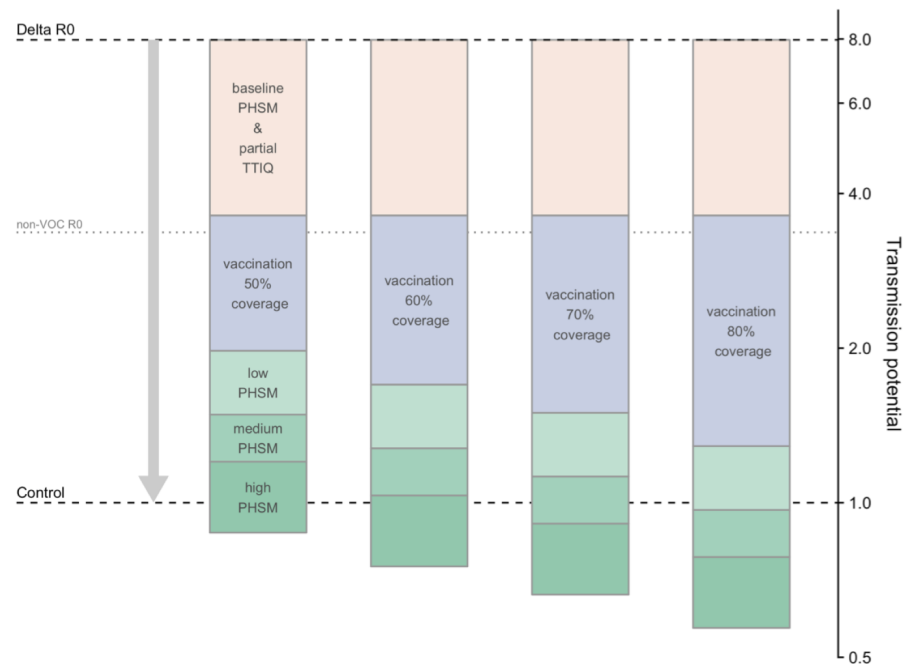
Data Analysis

All data analyses for this work were carried out in R [31]. R code to reproduce these analyses are available at <https://github.com/aus-covid-modelling/NationalCabinetModelling>. This code uses outputs from regular situational assessment work [19, 18] that are conducted using data provided under confidential agreement from the Australian Commonwealth Government (see also statement in Ethics), and that the authors are not authorised to make available. Code to create figures 3 and S1 can be found in [32].

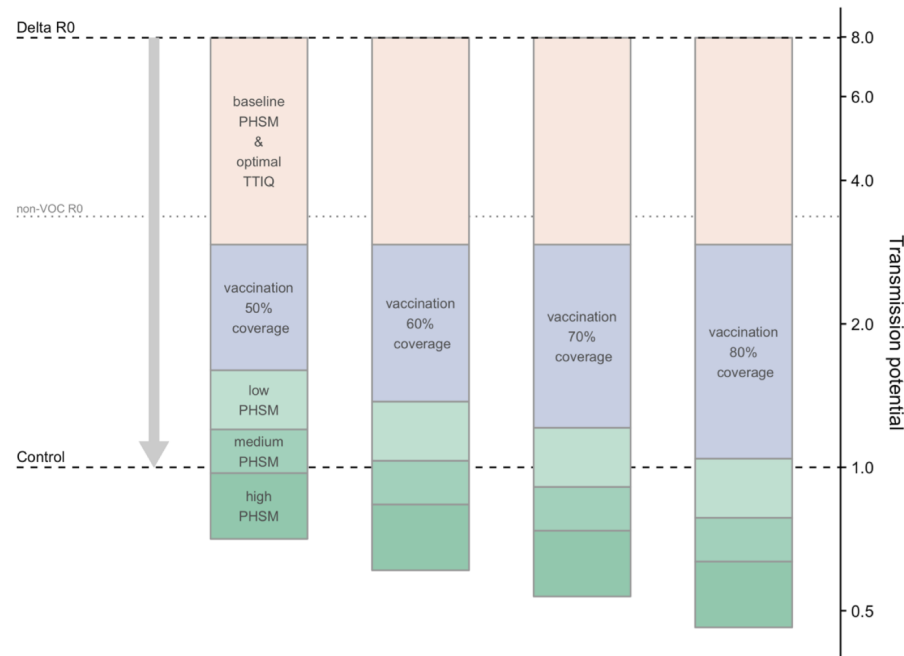
RESULTS

From an R_0 of 8 for the Delta variant, with baseline PHSMs and partial TTIQ in place TP is reduced to 3.6, which serves as a baseline to which other interventions are added 1. The effects of vaccination and more stringent PHSMs on TP are mathematically multiplicative, so the results displayed in figure 1 use a logarithmic y-axis in order to easily see the relative magnitude of each intervention. The results demonstrate that as vaccination coverage increases, less stringent PHSMs are required to bring TP below 1 and thus control epidemic activity (figure 1). Maintaining a rapid and highly effective TTIQ response capacity is critical for ongoing epidemic control. Should TTIQ responses become only partially effective due to high caseloads, high PHSM would be needed to curb transmission at the 50% and 60% coverage thresholds, whilst low PHSMs may be sufficient for control at 80% coverage (figure 1). More optimistically, the combination of 70% vaccine coverage and ongoing low PHSMs would likely be sufficient for control, if optimal TTIQ can be maintained (figure 1). Note that compliance with imposed measures will vary their effectiveness between populations and time-points. This uncertainty is conceptually represented by the upper and lower bounds of each 'box' for each set of restrictions in figure 1.

The choice of age-structured vaccine allocation strategy has a slight effect on Transmission Potential, though this varies with level of vaccine coverage (table 5). The contribution varies considerably by age-group due to differential mixing rates (figure 2, and figures S3, S4, and S5), and while a transmission



(a) Partial TTIQ.



(b) Optimal TTIQ.

Figure 1. Combined effects of vaccination and PHSM scenarios on COVID-19 transmission potential under the ‘Transmission reducing’ vaccination scenario assuming only partial (a) or optimal (b) TTIQ effectiveness, due to high caseloads. (NB the logarithmic scale of the y-axis enables comparison of the multiplicative effect sizes of these measures without depending on the order in which they are plotted.)

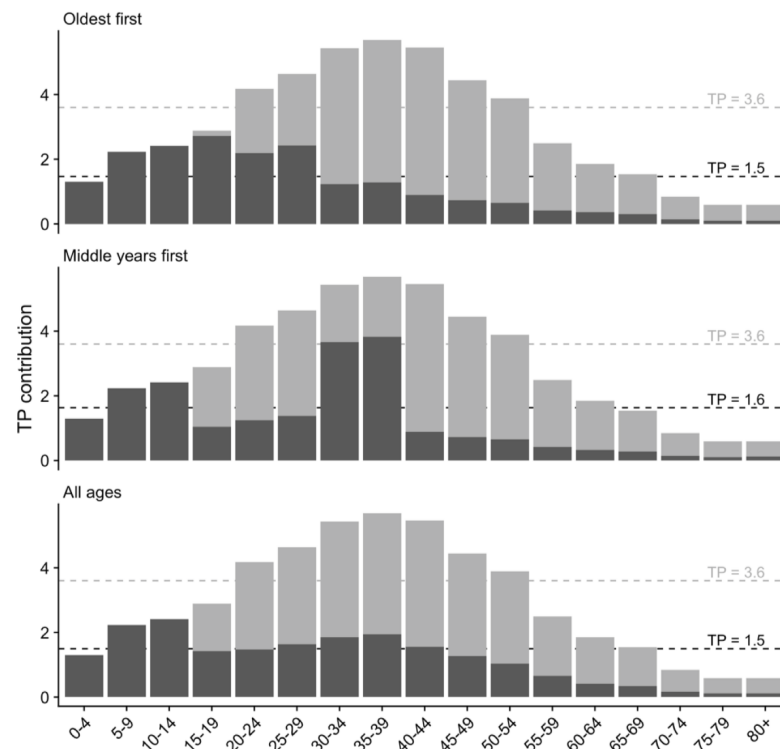


Figure 2. Impact of the four different vaccine allocation strategies on TP by age category, resulting in the overall TP achieved by 70% age eligible population coverage. Dark grey represents TP contribution after vaccination, and light grey in the absence of vaccination. Other coverage levels (50, 60 and 80% are in figures S2a-c). Dashed lines correspond to whole population differences in TP with age-group interactions included.)

reducing allocation strategy tends to best reduce TP as intended the effect is slight, and the advantage depends on the overall coverage level of vaccination. Vaccinating the 40+ years first tended to perform worst at lower levels of coverage, requiring the largest proportion of time under strict PHSMs, however this became unimportant at higher levels of vaccination.

Tables 6 and 7 compare the proportion of time that would need to be spent with high PHSM on top of ongoing light restrictions to maintain case counts at some level, by vaccine coverage and allocation strategy. We assume periodic switching between low PHSM and high PHSM over a long period with the same vaccination coverage. With long-term coverage held at 50%, 60%, or 70%, high PHSM would be needed for significant fractions of time (18-89%) if caseloads escalate, leading to 'partial' TTIQ effectiveness. For the 'optimal' TTIQ scenario and an achieved adult population coverage of 70%, high PHSM would be needed rarely if at all.

Figure 3 represents the proportion of time spent under each of the restriction stringency settings

Table 5. Scaled values of Delta variant transmission potential (TP) for 50%, 60%, 70% and 80% population vaccination coverage for each allocation strategy, with vaccines delivered per 4. We use a baseline TP of 3.6, which corresponds to TP under baseline PHSMs and partial TTIQ.

Allocation strategy	Eligible population coverage (16+)			
	50%	60%	70%	80%
Oldest first	2.1	1.7	1.5	1.3
40+ years first	2.1	1.9	1.6	1.3
All adults	2	1.7	1.5	1.3
Transmission reducing	2.1	1.8	1.4	1.3

Table 6. Percentage of time high PHSM would need to be in place for long-term control, with reversion to low PHSM at other times, for 50%, 60%, 70% and 80% population coverage achieved under the three age-based allocation strategies. These scenarios assume partial TTIQ effectiveness, under high caseloads. Standard age (60+) and dosing interval (12 weeks) recommendations are assumed for AZ vaccine.

Allocation strategy	Eligible population coverage (16+)			
	50%	60%	70%	80%
Oldest first	82%	49%	18%	0%
40+ years first	89%	67%	39%	2%
All adults	75%	46%	22%	0%
Transmission reducing	87%	52%	10%	0%

Table 7. Percentage of time high PHSM would need to be in place for long-term control, with reversion to low PHSM at other times, for 50%, 60%, 70% and 80% population coverage achieved under the three age-based allocation strategies. These scenarios assume optimal TTIQ effectiveness, under high caseloads. Standard age (60+) and dosing interval (12 weeks) recommendations are assumed for AZ vaccine.

Allocation strategy	Eligible population coverage (16+)			
	50%	60%	70%	80%
Oldest first	42%	9%	0%	0%
40+ years first	49%	27%	0%	0%
All adults	35%	6%	0%	0%
Transmission reducing	47%	12%	0%	0%

for different levels of vaccine coverage and intensity of case finding and management strategies, with corresponding costs of these restrictions as estimated by Treasury shown on the right of the figure [13]. These combined outputs demonstrate the substantive cost savings associated with avoidance of lockdown and provided additional justification for delaying reopening until achieving 70% threshold vaccine coverage. The overlay of some degree of social measures at this threshold supported transmission reduction and helped to maintain the effectiveness of an active case finding strategy focused on minimising health impacts. At 80% coverage these restrictions could be eased without any envisaged lockdown requirement.

DISCUSSION

This work formed the basis of a significant contribution to easing of COVID-19 restrictions in Australia in late 2021 by providing a structured framework to explore the effect of management decisions on transmission of SARS-CoV-2 in Australia based on contemporary evidence [13, 14, 15, 16]. Specifically, it covers the pandemic period up to mid-late 2021, during which time the Delta variant became established in several eastern Australian states resulting in imposition of stringent lockdowns to manage transmission and clinical burden, and closure of interstate borders. For those states, our recommendation of a target vaccination level of at least 70% was a pathway out of restrictions and a way to reconnect with 'COVID-zero' jurisdictions. In the other states, high vaccine coverage provided confidence that COVID-19 impacts could be mitigated sufficiently to avoid substantive health system and social disruption, and at 80% coverage only minimal social and behavioural measures would be required to support vaccination.

The scenarios in this study representing a single national COVID-19 epidemic were clearly (and deliberately) artificial and served to inform high level policy strategy. Beyond defining threshold vaccine targets, they highlighted the importance of a combination of timely public health responses (TTIQ) and ongoing social and behavioural measures (PHSMs) to constrain SARS-CoV-2 transmission. It was recognised that at high caseloads, maintenance of optimal TTIQ was likely infeasible. In such instances, jurisdictions retained the flexibility to strengthen PHSMs generally or locally (as envisaged in the National Plan [12]) to regain local epidemic control. The required intensity and duration of measures was to be informed by ongoing situational assessment of transmission and its related health impacts [19].

In August 2021 when these analyses were first reported, evidence of vaccine efficacy against acquisition and onward spread of infection had raised hopes that equitable vaccine distribution could substantially limit global transmission and burden of COVID-19 disease [33]. However, studies undertaken by multiple

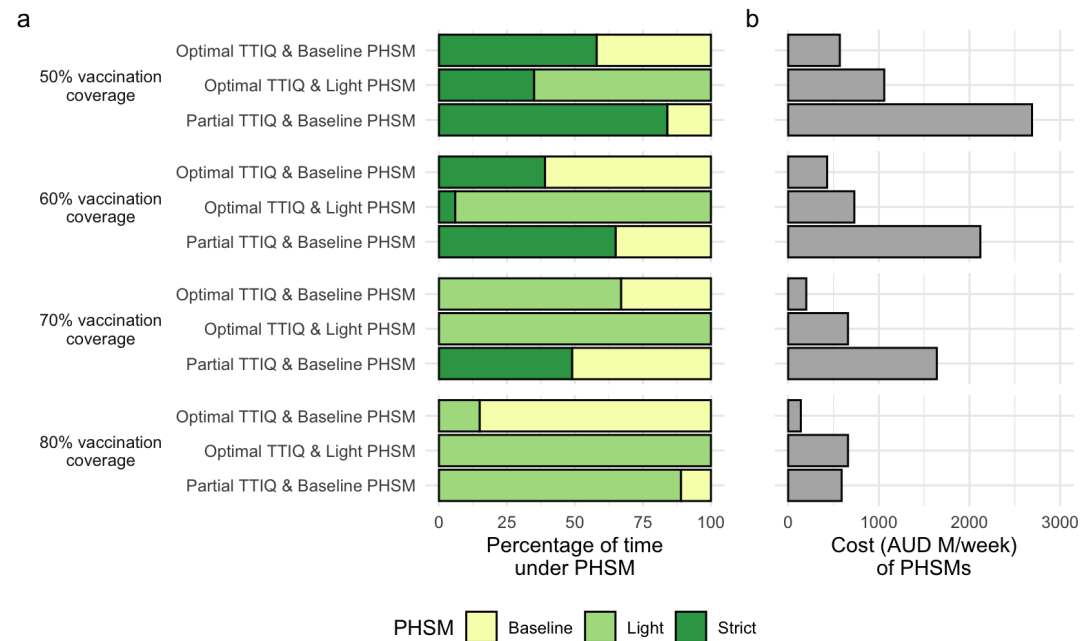


Figure 3. Percentage of time necessary under PHSMs to constrain outbreaks (a) and cost (in millions of Australian dollars per week) of the least expensive PHSM bundle able to constrain outbreaks (b) under each combination of TTIQ effectiveness and ongoing PHSMs, at 50, 60, 70, and 80% vaccination coverage of the 16+ adult population, and an "All adults" vaccine allocation scenario. Data in (a) from S2, S3, S4, and S5, and (b) from [13]

modelling groups supporting decision making to ease UK lockdowns during the Delta era similarly cautioned against the lifting of all restrictions ('freedom day') following achievement of vaccine targets, given imperfect vaccine protection and potential for resurgence [34]. In support of this position, a rapid rebound of Delta infections following reopening in the Netherlands had necessitated re-imposition of social measures within only weeks of easing. At that time, global concern about driving emergence of further variants remained high and strong suppression strategies were favoured by the World Health Organisation [35].

Our findings were aligned with and benchmarked against the UK modelling reports, including the recommendation for ongoing social constraints, but were novel in two main aspects. For the majority of Australian jurisdictions, relaxation of border restrictions would allow importation of infections into 'COVID-zero' settings, requiring high confidence in model recommendations. The use of transmission potential as a novel metric enabled anticipation of case loads in settings where there were presently none [10]. Given ongoing global discussions about competing health and economic impacts of COVID-19 and social measures for its control [36], the accompanying Treasury analyses were highly influential in whole of government decision making. We are unaware of similar publicly available estimates of the costs of PHSMs under different levels of vaccine coverage from other country settings.

The model assumed fixed efficacy of vaccine protection against transmission and disease at a single time point and did not incorporate waning immunity. At the time of analysis, longevity of population experience with COVID-19 vaccines was limited and evidence of the rate of loss of protection was sparse. Israel was the first country to approve booster doses on July 30 2021, initially for individuals aged 60 and above. This recommendation was prompted by observation of breakthrough infections following primary vaccination consistent with waning immunity [37]. Booster requirements were then a contentious issue, with many arguing that broad provision of a third vaccine dose was unnecessary in highly immunised populations. There were particular concerns about impacts on supply for global vaccine equity, given persistently low primary vaccine coverage and access in many low and middle income countries [38].

Note that the visualisations and metrics presented in this paper were used to evaluate the general

viability of different suites of measures, under a very uncertain future epidemiological situation and period of time. These metrics were not sufficient to calculate the likely morbidity and mortality outcomes of COVID-19 under specific rollout strategies and changes given the dynamic nature of vaccination and transmission. Related work extended our initial findings on transmission potential into an agent based model framework to estimate those impacts and is reported elsewhere [39].

It was further recognised that the national COVID-19 epidemic had been, and would continue to be, a ‘fire’ fought on multiple fronts across Australia’s geographically distributed population, largely concentrated in coastal urban cities. We recommended that particular attention be paid to groups in whom socioeconomic, cultural and other determinants were anticipated to result in higher transmission and/or disease outcomes. In addition, achievements of vaccination targets at small area level was critical to ensure equity of program impact, as ongoing outbreaks in under-vaccinated populations were considered likely, and would need to be supported by optimisation of localised public health responses. These issues were the focus of subsequent work [11, 39].

In reality, the applicability of these defined thresholds to reopening goals was made redundant by importation and rapid transmission of the Omicron SARS-CoV-2 variant in December 2021. Dissemination of this and subsequent immune escape variants has led to global reconsideration of the role of vaccination in the control of COVID-19. Moreover, vaccine protection against severe disease appears more robust and sustained than that against transmission, particularly in the context of waning post-immunisation neutralising antibody titres (REF). This emerging understanding has reoriented strategic vaccine use towards promoting population resilience against severe disease outcomes, rather than transmission reduction. Subsequent work focuses on the implications of variant emergence for deployment of vaccines and other control measures in the era of Omicron and beyond.

ETHICS

The study was undertaken as urgent public health action to support Australia’s COVID-19 pandemic response. The study used data from the Australian National Notifiable Disease Surveillance System (NNDSS) provided to the Australian Government Department of Health under the National Health Security Agreement for the purposes of national communicable disease surveillance. Data from the NNDSS were supplied after de-identification to the investigator team for the purposes of provision of epidemiological advice to government. Contractual obligations established strict data protection protocols agreed between the University of Melbourne and sub-contractors and the Australian Government Department of Health, with oversight and approval for use in supporting Australia’s pandemic response and for publication provided by the data custodians represented by the Communicable Diseases Network of Australia. The ethics of the use of these data for these purposes, including publication, was agreed by the Department of Health with the Communicable Diseases Network of Australia.

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540 SUPPLEMENTARY MATERIALS

541 Fraction of time in lockdown

542 Where a vaccination scenario leads to either a $TP_1 > 1$ with one PHSM bundle and $TP_2 < 1$ with a more
 543 stringent bundle, the long-term average TP can be maintained at 1, i.e. with daily case counts neither
 544 growing nor shrinking over the long term, by alternating between the two PHSM bundle states. Whilst
 545 the first PHSM bundle is in place cases will grow, and whilst the more stringent bundle is in place cases
 546 will shrink, leading to an oscillation of case counts around some average level S1. This reflects a strategy
 547 that might be used to keep cases below a health sector capacity limit in the event that there is long-term
 548 community transmission and under the necessary simplifying assumption that vaccination coverage is
 549 static.

Where $TP_1 > 1$ and $TP_2 < 1$, we can calculate the fraction of time spent under more stringent PHSMs
 as:

$$fraction = -\frac{\log(TP_1)}{\log(TP_2) - \log(TP_1)} \quad (S1)$$

550 Where $TP_1 < 1$ the fraction is zero, and as the more stringent PHSM bundle is not needed. Where $TP_2 > 1$
 551 no fraction exists, because even the more stringent PHSM bundle could not control transmission.

552 Note that this fraction of time under more stringent PHSMs is independent of the sequence or duration
 553 of the periods under more stringent restrictions; a strategy of rapid switching on and off of restrictions, or
 554 one of alternating long periods with or without restrictions would both lead to long-term maintenance
 555 of $TP = 1$, provided the fraction of time in each condition is the same and that daily case numbers are
 556 either growing or contracting exponentially (i.e., no significant susceptible depletion alters the growth
 557 rate), e.g. S1. However for practical reasons, a rapid switching between states is unlikely to be used. The
 558 overall number of cases (as opposed to the long-term average TP) will however be dependant on the rate
 559 of switching, but it is assumed in all calculations in this paper that switching is sufficient to maintain total
 560 case numbers below a level which would result in overwhelming of the public healthcare system.

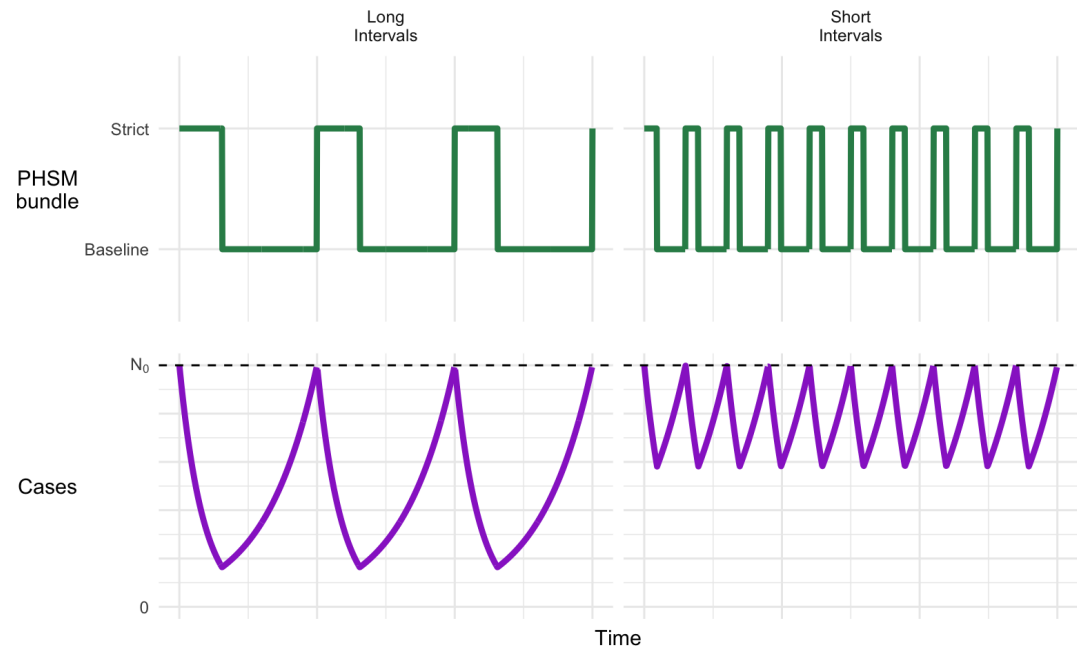


Figure S1. Example of how daily cases may fluctuate over time in response to stringency of PHSMs using 80% vaccination coverage, partial TTIQ and reversion to baseline PHSMs, with either long or short intervals between switching PHSMs. In this example, to maintain a stable $TP \leq 1$ over time requires strict PHSMs for $\geq 31\%$ of the time 3, S2. Starting with strict PHSMs as here results in cases fluctuating below the initial daily case number (N_0 , dotted line). The length of intervals under respective PHSMs (left vs. right) will not alter the overall average TP (though may affect total case numbers).

Transmission matrix

Population mixing within and between age groups is configured based on widely accepted social contact matrices published by [29]. It has been expanded to include an 80+ age class (assumed to have the same mixing rates as 75-79 years) S2. Age-specific susceptibility and transmissibility estimates from [30] are used and transmission rates have been calibrated to our baseline population-wide TP of 3.6.

The greatest mixing intensities are anticipated between individuals aged from 15-24 years, remaining high through adults of working age. While intense school-based mixing is anticipated between children aged 5-14, the transmission matrix accounts for the relatively low observed infectiousness of this age group, associated with a high proportion of asymptomatic infections.

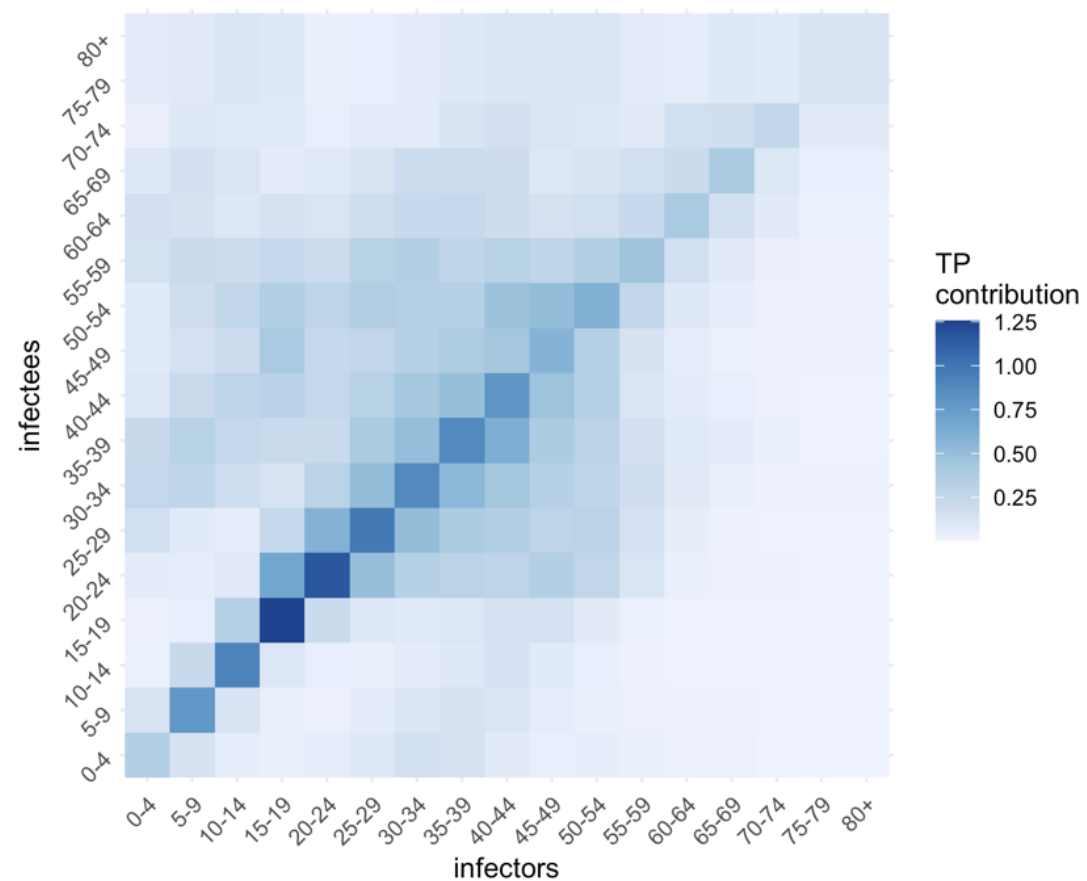


Figure S2. Age-based transmission matrix derived from [29].

570 **Supplementary figures and tables**

Table S1. Detailed description of measures implemented under PHSM ‘bundles’.

	High PHSM	Medium PHSM	Low PHSM	Baseline PHSM
Reference period	VIC 23 August 2020	NSW 1 July 2021	NSW 23 August 2020	NSW March 2021
Stay at home orders	Stay-at-home except essential purposes	Stay-at-home except for work, study and essential purposes	No stay-at-home orders	No stay-at-home orders
Density restrictions	4 m ² rule	2 m ² rule	2 m ² rule	2 m ² rule
Retail trade	Non-essential retailers and venues closed to public. Take away and home delivery only	Increased retail activity, subject to density restrictions. Seated dining for small groups at cafes/restaurants	Social distancing rules apply. Larger groups allowed.	Social distancing rules apply
Work	Only workplaces categorised as permitted work allowed to operate on-site and subject to restrictions	Work from home if possible, capacity limits and restrictions on office space apply	Return to work, but social distancing and capacity restrictions on office space apply	1.5 m ² rule
Schools and childcare	Closed – remote learning only	Closed or graduated return	Open	Open
Capacity restrictions	No gatherings - Non-essential venues etc closed.	Indoor venues closed. Capacity limits restricted to small groups outdoors	Recreational activities allowed and venues open but social distancing and capacity limits apply	Large sporting venues to operate at 70 per cent capacity
Travel restrictions	Essential movements only within 5 or 10 km radius. No intra- or inter-state travel	Non-essential travel limited. No intra- or inter-state travel	No travel restrictions. Interstate travel allowed	No travel restrictions. Interstate travel allowed
Other	Curfew. No household visitors and 2-person limit on exercise	5 visitors to household and limited outdoor gatherings e.g., 10 people	Requirements for record keeping, COVID-safe plans etc.	

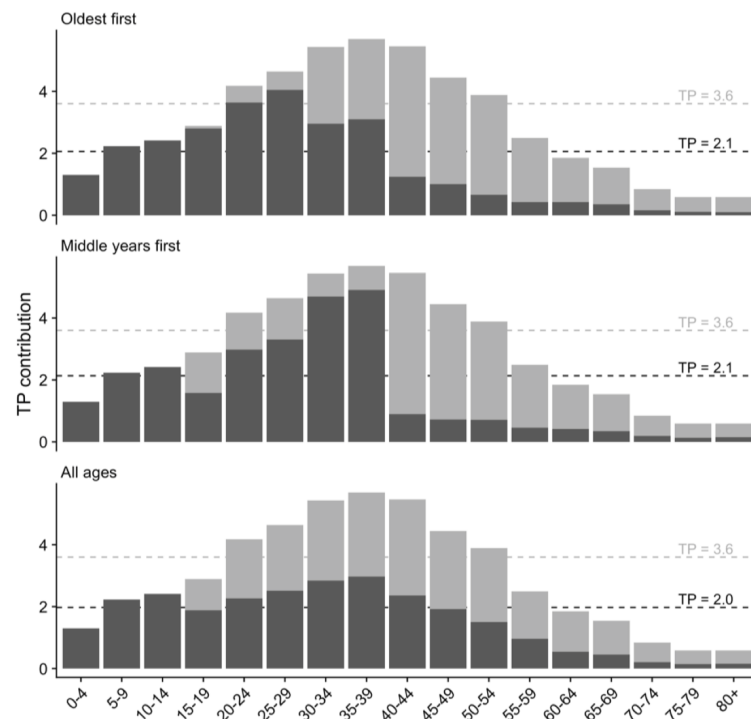


Figure S3. mpact of the four different vaccine allocation strategies on TP by age category, resulting in the overall TP achieved by 50% age eligible population coverage. Dark grey represents TP contribution after vaccination, and light grey in the absence of vaccination. Dashed lines correspond to whole population differences in TP with age-group interactions included (is marginal contribution of each age-group / column sums of transmission matrix)

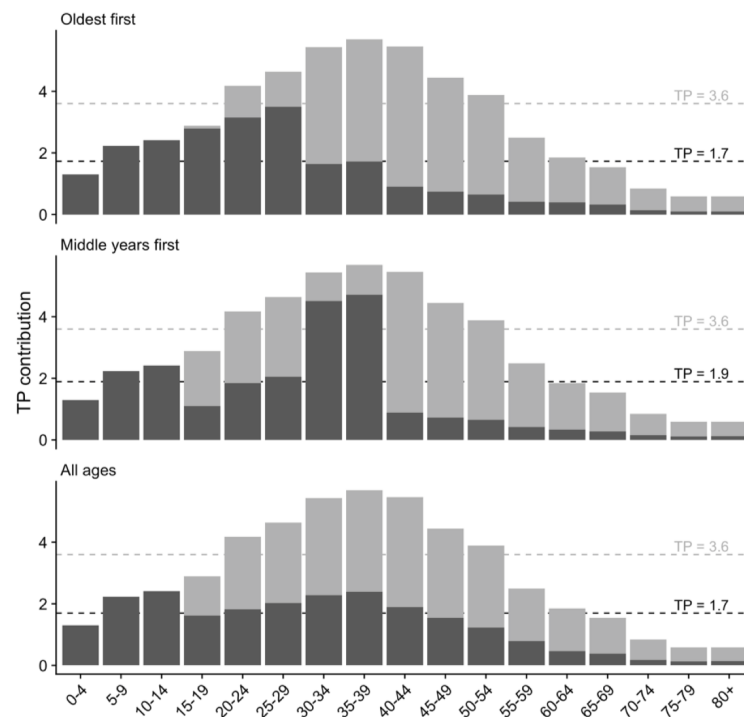


Figure S4. mpact of the four different vaccine allocation strategies on TP by age category, resulting in the overall TP achieved by 60% age eligible population coverage. Dark grey represents TP contribution after vaccination, and light grey in the absence of vaccination. Other coverage levels (50, 60 and 80% are in figures S2a-c). Dashed lines correspond to whole population differences in TP with age-group interactions included (is marginal contribution of each age-group / column sums of transmission matrix)

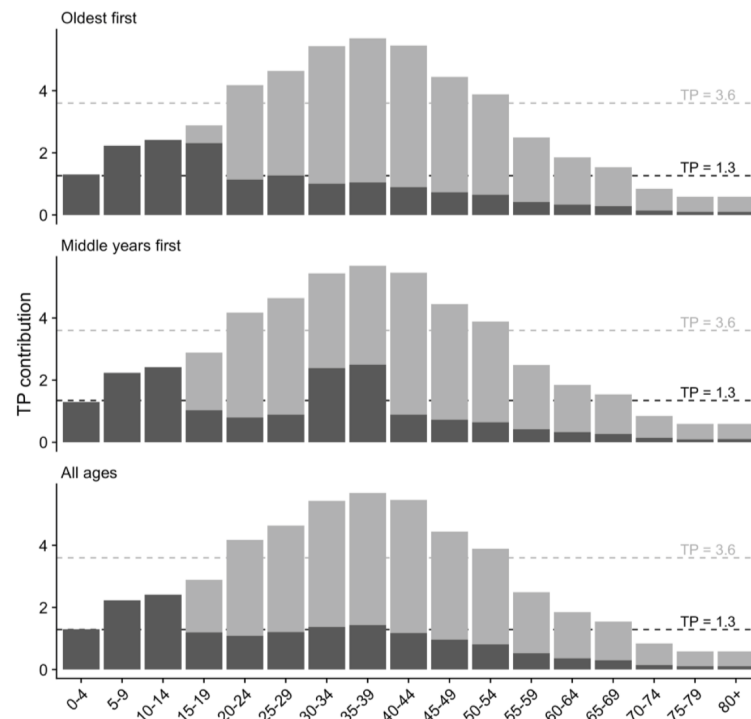


Figure S5. mpact of the four different vaccine allocation strategies on TP by age category, resulting in the overall TP achieved by 80% age eligible population coverage. Dark grey represents TP contribution after vaccination, and light grey in the absence of vaccination. Dashed lines correspond to whole population differences in TP with age-group interactions included (is marginal contribution of each age-group / column sums of transmission matrix)

Table S2. Proportion of time lockdowns are needed to constrain transmission when the TTIQ public health response is only partially effective, due to high caseloads

Vaccine coverage	Allocation Scenario	Light restrictions only	Moderate lockdowns only	Strict lockdowns only
50%	Oldest first	Not possible to constrain outbreak	Not possible to constrain outbreak	89%
	40+ years first	Not possible to constrain outbreak	Not possible to constrain outbreak	93%
	All adults	Not possible to constrain outbreak	Not possible to constrain outbreak	84%
60%	Oldest first	Not possible to constrain outbreak	Not possible to constrain outbreak	67%
	40+ years first	Not possible to constrain outbreak	Not possible to constrain outbreak	78%
	All adults	Not possible to constrain outbreak	Not possible to constrain outbreak	65%
70%	Oldest first	Not possible to constrain outbreak	77%	47%
	40+ years first	Not possible to constrain outbreak	99%	60%
	All adults	Not possible to constrain outbreak	81%	49%
80%	Oldest first	82%	47%	29%
	40+ years first	Not possible to constrain outbreak	59%	36%
	All adults	89%	51%	31%

Table S3. Proportion of time lockdowns are needed to constrain transmission when the TTIQ public health response is optimally effective.

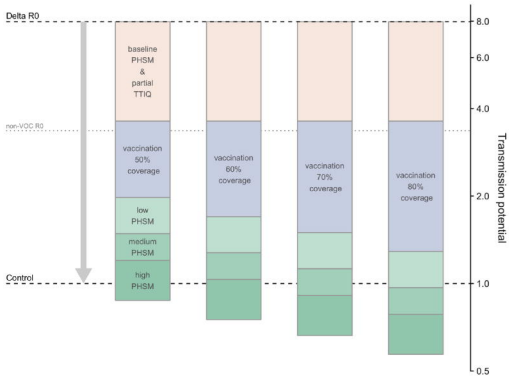
Vaccine coverage	Allocation Scenario	Light restrictions only	Moderate lockdowns only	Strict lockdowns only
50%	Oldest first	Not possible to constrain outbreak	Not possible to constrain outbreak	63%
	40+ years first	Not possible to constrain outbreak	Not possible to constrain outbreak	67%
	All adults	Not possible to constrain outbreak	94%	58%
60%	Oldest first	Not possible to constrain outbreak	67%	41%
	40+ years first	Not possible to constrain outbreak	86%	52%
	All adults	Not possible to constrain outbreak	64%	39%
70%	Oldest first	60%	34%	21%
	40+ years first	97%	56%	34%
	All adults	67%	38%	23%
80%	Oldest first	7%	4%	3%
	40+ years first	29%	17%	10%
	All adults	15%	8%	5%

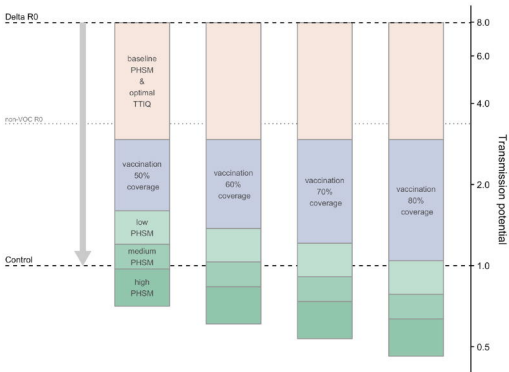
Table S4. Proportion of time lockdowns needed to constrain transmission when the TTIQ public health response is only partially effective, due to high caseloads, and where light restrictions are always in place.

Vaccine coverage	Allocation Scenario	Moderate lockdowns only	Strict lockdowns only
50%	Oldest first	Not possible to constrain outbreak	82%
	40+ years first	Not possible to constrain outbreak	89%
	All adults	Not possible to constrain outbreak	75%
60%	Oldest first	Not possible to constrain outbreak	49%
	40+ years first	Not possible to constrain outbreak	67%
	All adults	Not possible to constrain outbreak	46%
70%	Oldest first	46%	18%
	40+ years first	97%	39%
	All adults	55%	22%
80%	Oldest first	0%	0%
	40+ years first	4%	2%
	All adults	0%	0%

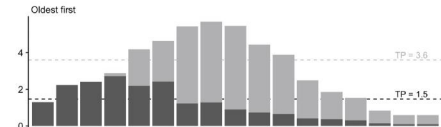
Table S5. Proportion of time lockdowns needed to constrain transmission when the TTIQ public health response is optimally effective and where light restrictions are always in place.

Vaccine coverage	Allocation Scenario	Moderate lockdowns only	Strict lockdowns only
50%	Oldest first	Not possible to constrain outbreak	42%
	40+ years first	Not possible to constrain outbreak	49%
	All adults	87%	35%
60%	Oldest first	23%	9%
	40+ years first	66%	27%
	All adults	15%	6%
70%	Oldest first	0%	0%
	40+ years first	0%	0%
	All adults	0%	0%
80%	Oldest first	0%	0%
	40+ years first	0%	0%
	All adults	0%	0%

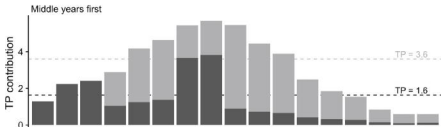




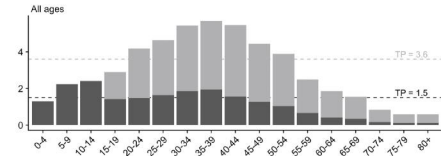
Oldest first



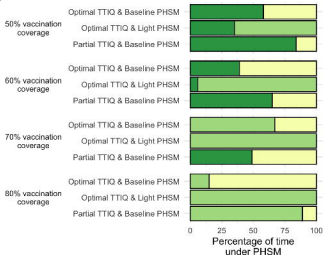
Middle years first



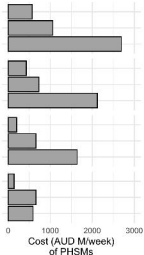
All ages



a



b



PHSM Baseline Light Strict

